

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I Jeevakrishna. R (192511108) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Jeevakrishna. R

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

1. Event Communication Security (Caesar Cipher)

a. Encrypt "MEET AT NOON" using a shift of 4

In a Caesar Cipher, every letter is shifted forward by a fixed number of positions.

Plaintext M E E T A T N O O N

Shift +4 Q I I X E X R S S R

Ciphertext:

QIIX EX RSSR

b. How can the receiver decrypt it?

The authorized receiver knows that the sender used a shift of 4.

They simply shift every ciphertext letter **4 positions backward**.

Example:

- $Q \rightarrow M$
- $I \rightarrow E$
- $X \rightarrow T$
- $R \rightarrow N$

Thus,

QIIX EX RSSR $\rightarrow$ MEET AT NOON

This works because Caesar Cipher uses the same fixed shift for encryption and decryption.

2. Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

a. Determine the original message

This is encrypted using the **Atbash Cipher**, where:

$A \leftrightarrow Z$

$B \leftrightarrow Y$

$C \leftrightarrow X$

...

Applying the reverse mapping:

$G \rightarrow T$

$S \rightarrow H$

$R \rightarrow I$

$H \rightarrow S$

$RH \rightarrow IS$

$Z \rightarrow A$

$HVXIVG \rightarrow SECRET$

Original Message:

THIS IS A SECRET

b. Justification

The message follows characteristics of the Atbash cipher:

- Same letters always map to the same ciphertext.
- Word lengths remain unchanged.
- No key is required.
- It is a monoalphabetic substitution cipher.
- The phrase "THIS IS A SECRET" is a common plaintext matching the observed letter patterns.

Therefore, the intercepted text is most likely encrypted using the Atbash substitution cipher.

3. Secure Messaging System (Vigenère Cipher)

Plaintext:

HELLO

Keyword:

KEY

a. Step-by-step encryption

Repeat the keyword until it matches the plaintext length.

Plaintext H E L L O

Keyword K E Y K E

Convert letters into numbers:

Letter Value

H	7
E	4
L	11
L	11
O	14
K	10
E	4
Y	24
K	10
E	4

Now add modulo 26.

Plain Key Total Cipher

7	10	17	R
4	4	8	I
11	24	35→9	J
11	10	21	V
14	4	18	S

Ciphertext**RIJVS****b. Role of the repeating keyword**

The keyword repeats continuously.

KEY becomes

KEYKE

Each keyword letter shifts the corresponding plaintext letter differently.

Because every letter is encrypted with a different shift, Vigenère Cipher is much stronger than Caesar Cipher and is harder to break using simple frequency analysis.

4. Suspicious Digital Asset Investigation (Steganography)

a. How can investigators detect hidden data?

Investigators can use several methods:

1. Statistical Analysis

Checks unusual pixel patterns that may indicate hidden information.

2. Steganalysis Software

Tools like:

- StegExpose
- Stegdetect
- OpenStego
- zsteg

can identify hidden messages inside images.

3. Compare Original and Suspected Image

Differences in pixel values may reveal hidden data.

4. LSB (Least Significant Bit) Analysis

Many steganography methods hide information by modifying the least significant bits of pixels.

Investigators inspect these bits to recover hidden information.

b. Two techniques to improve security

1. Encrypt the secret message before hiding it

Even if extracted, the attacker cannot read the data without the encryption key.

2. Random Pixel Embedding

Instead of storing data sequentially, use a secret key to randomly choose pixel locations.

This makes detection significantly more difficult.

5. Legacy Encryption in Corporate Communication

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

a. Recover the original message

The ciphertext was encrypted using a **3-Rail Rail Fence Cipher**.

Writing the ciphertext across three rails and reconstructing the zig-zag pattern gives:

WE ARE DISCOVERED FLEE AT ONCE

b. Why is this method vulnerable?

The Rail Fence Cipher is insecure because:

- It only rearranges letters without changing them.
- Letter frequencies remain unchanged.
- It has a small key space (few rail values).
- Modern computers can brute-force every possible rail count within milliseconds.
- It provides no protection against cryptanalysis.

Therefore, it is unsuitable for protecting confidential information in modern communication systems.

c. One practical enhancement

Instead of using only Rail Fence Cipher:

- Encrypt the message using **AES (Advanced Encryption Standard)** before transmission.

Advantages:

- Strong 128/192/256-bit encryption
- Resistant to brute-force attacks
- Widely used in banking, government, and secure communications
- Provides much higher confidentiality than classical transposition methods

Conclusion

This assessment demonstrates the practical application of classical cryptographic techniques, including Caesar Cipher, Atbash Cipher, Vigenère Cipher, Steganography, and Rail Fence Cipher. It also highlights why these traditional methods are insufficient for modern security and why stronger algorithms such as AES are preferred today.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation